Advances in the Equivariant Minimal Model Program and Their Applications in Complex and Arithmetic Dynamics

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Abstract This note reports some advances in the Equivariant Minimal Model Program (EMMP) for non-isomorphic surjective endomorphisms and their applications in complex and arithmetic dynamics.

1 Introduction

We work over an algebraically closed field of characteristic 0 unless otherwise specified. A fair amount is known in any characteristic and we refer to [11] and its references for more information.

Consider a finite sequence of dominant rational maps of projective varieties

$$(*): \quad X_1 \dashrightarrow X_2 \dashrightarrow \cdots \dashrightarrow X_r.$$

Let $f : X_1 \to X_1$ be a surjective endomorphism. We say the sequence $(*)$ is *f-equivariant* if there exists the following commutative diagram

$$\begin{array}{ccccccc} X_1 & - - \succ & X_2 & - - \succ & \cdots & - - \succ & X_r \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & & & \downarrow{\scriptstyle f_r} \\ X_1 & - - \succ & X_2 & - - \succ & \cdots & - - \succ & X_r \end{array}$$

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where $f_1 = f$ and all the f_i are surjective endomorphisms. Let S be a set of surjective endomorphisms of X. We say the sequence $(*)$ is *S-equivariant* if it is g-equivariant for every $g \in S$.

The first part of this paper focuses on the equivariance of the Minimal Model Program (MMP), i.e., how to construct an Equivariant Minimal Model Program (EMMP); see Sect. 2. In general, EMMP holds only for certain kinds of endomorphisms, say satisfying property $\mathcal{P}$. One key task is about the completeness of the EMMP: whether the property $\mathcal{P}$ can be preserved by EMMP. There are two important situations: polarized and int-amplified. We will give several different criteria for these endomorphisms in Sect. 3. With these criteria, we can show that polarized and int-amplified endomorphisms can be preserved under EMMP.

We here briefly summarize several results on the int-ampified EMMP in the following theorem. It will be explained in detail in Sect. 4 where the behaviour of the MMP will be discussed.

Theorem 1 *Let X be a normal projective variety admitting an int-amplified endomorphism. Then we have:*

(1) Suppose (X, Δ) is lc for some effective $\mathbb{R}$-divisor Δ. Then there are only finitely many $(K_X + \Delta)$-negative extremal rays.
(2) Suppose X is $\mathbb{Q}$-factorial. Then any finite sequence of MMP starting from X is G-equivariant for some finite-index submonoid G of $\mathrm{SEnd}(X)$, where $\mathrm{SEnd}(X)$ is the monoid of all surjective endomorphisms of X.
(3) Suppose X is $\mathbb{Q}$-factorial and klt. Then there any MMP can end up with a Q-abelian variety, i.e., a quasi-étale quotient of an abelian variety.

We introduce several special situations when EMMP holds.

- X is a normal projective surface admitting a non-isomorphic surjective endomorphism; see Sect. 5.
- X is a smooth projective threefold with the Kodaira dimension $\kappa(X) \geq 0$ and f is non-isomorphic; see Sect. 6.
- X is klt Calabi-Yau type and f is quasi-amplified; see Sect. 7.

The second part of this paper will focus on applications of EMMP to several problems and conjectures in complex and arithmetic dynamics.

When X is a Fano manifold, EMMP naturally holds since the nef cone is a rational polyhedron. In this situation, we wish to use EMMP to characterize the structure of X. Fakhruddin conjectured that X has to be toric if it admits a polarized endomorphism. This is in general quite hard and what we know so far is the following result; see Sect. 8.

Theorem 2 *Let X be a smooth Fano threefold. Then the following are equivalent.*

(1) X is a product of $\mathbb{P}^1$ and a del Pezzo surface.
(2) X admits a non-isomorphic surjective endomorphism which is not polarized even after iteration.

(3) X admits a non-isomorphic surjective endomorphism which is non-amplified (or equivalently, non-int-amplified).

Broustet and Gongyo conjectured that a normal projective variety admitting a polarized endomorphism is log Calabi-Yau. The current progress toward this conjecture is the following; see Sect. 9.

Theorem 3 *Let $f : X \to X$ be a polarized endomorphism of a normal projective variety X. Suppose either* $\dim X \leq 2$ *or X is a smooth projective threefold. Then X is of Calabi-Yau type.*

In the last two Sects. 10 and 11, we will survey current progress towards the Kawaguchi-Silverman conjecture and the Zariski dense orbit conjecture, mainly from the EMMP aspect.

We will outline the ideas while referring to the original papers for detailed proofs.

2 The Minimal Model Program (MMP)

In this section, we briefly introduce the classical Minimal Model Program (MMP). We begin with some terminology.

Setup 1 Let X be a projective variety of dimension n. Denote by $\mathrm{Pic}(X)$ the group of Cartier divisors modulo linear equivalence and $\mathrm{Pic}^0(X)$ the subgroup of the classes in $\mathrm{Pic}(X)$ which are algebraically equivalent to 0. Let $\mathrm{NS}(X) = \mathrm{Pic}(X)/\mathrm{Pic}^0(X)$ be the *Néron-Severi group*. Denote by $\mathrm{N}^1(X) := \mathrm{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$. Let Z and Z' be two r-cycles with real coefficients. Z is said to be *numerically equivalent* to Z' (denoted by $Z \equiv Z'$) if $(Z - Z') \cdot H_1 \cdots H_r = 0$ for any Cartier divisors $H_1, \ldots, H_r$; it was called weakly numerically equivalent in our earlier papers and is weaker than the numerical equivalence defined in Fulton's book (cf. [16, Definition 19.1]). For Cartier divisors (regarded as $(\dim X - 1)$-cycles), the numerical equivalence here coincides with the usual one; see [75, Lemma 3.2]. Denote by $\mathrm{N}_r(X)$ the quotient vector space of r-cycles with real coefficients modulo the numerical equivalence. We refer to [39, Chap. II] and [21] for some basics of singularities of a variety or a pair, like, *lc (log canonical), klt (Kawamata log terminal), canonical, or terminal singularity*. A pair (X, Δ) is a normal variety X with a $\mathbb{Q}$-Cartier (boundary) divisor Δ such that all coefficients of Δ are in $[0, 1] \cap \mathbb{Q}$ and the adjoint divisor $K_X + \Delta$ is $\mathbb{Q}$-Cartier.

Let (X, Δ) be a pair with only lc singularities. Let $\pi : X \to S$ be a projective morphism to a variety S. We use $\mathrm{N}_1(X/S)$ to denote the space of 1-cycles generated by curves in fibres of π, modulo numerical equivalence. $\overline{\mathrm{NE}}(X/S)$ denotes the closure of the effective 1-cycles in $\mathrm{N}_1(X/S)$. The classical *cone theorem* asserts that

$$\overline{\mathrm{NE}}(X/S) = \overline{\mathrm{NE}}(X/S)_{(K_X+\Delta)\geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i]$$

where each $\mathbb{R}_{\geq 0}[C_i]$ is an *extremal ray* of $\overline{\mathrm{NE}}(X/S)$, generated by a rational curve C_i with $(K_X + \Delta) \cdot C_i < 0$. Moreover, for each C_i, the base point free theorem ensures a contraction morphism, with connected fibres, $\tau : X \to Y$ over S to a normal projective variety Y, called *the contraction of the* $(K_X + \Delta)$*-negative extremal ray* $\mathbb{R}_{\geq 0}[C_i]$, such that exactly the curves in the class of $\mathbb{R}_{\geq 0}[C_i]$ are contracted to points and one obtains the exact sequence

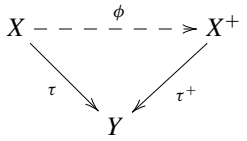
$$0 \to \mathrm{N}^1(Y) \xrightarrow{\tau^*} \mathrm{N}^1(X) \xrightarrow{\cdot C_i} \mathbb{R} \to 0.$$

In particular, we have the relation $(\dim_{\mathbb{R}} \mathrm{N}^1(X) =)\ \rho(X) = \rho(Y) + 1$ between the *Picard numbers*.

Assume further that (X, Δ) is a $\mathbb{Q}$-factorial log canonical pair. There are three cases for the contraction $\tau : X \to Y$ of a $(K_X + \Delta)$-negative extremal ray $\mathbb{R}_{\geq 0}[C]$.

Case 1 (Divisorial contraction). τ is a birational contraction with the exceptional locus (of points of X along which τ is not an isomorphism) being a prime divisor. Note that Y is still $\mathbb{Q}$-factorial with log canonical singularities.

Case 2 (Flipping contraction). τ is small, i.e., a birational contraction with the exceptional locus being of codimension ≥ 2 in X. In this case, Y is not $\mathbb{Q}$-factorial and we need to take the flipping diagram:



where τ^+ is a small projective birational morphism and

1. (X^+, Δ^+) is a $\mathbb{Q}$-factorial log canonical pair with $\Delta^+ = \phi_* \Delta$,
2. $K_{X^+} + \Delta^+$ is τ^+-ample, and
3. $\rho(X^+) = \rho(Y) + 1$.

We refer to [7, 28] for the existence of log canonical flips.

Case 3 (Fano contraction). This is the case where $\dim Y < \dim X$ (and general fibres are of Fano type). Note that Y is again $\mathbb{Q}$-factorial with log canonical singularities, see [21].

Remark 4 We refer to [22, Chapter 4] for the non-$\mathbb{Q}$-factorial Minimal Model Program.

Remark 5 There are two very hard conjectures in the Minimal Model Program. One is the termination of flips asserting that one can not have infinitely many flips when running a Minimal Model Program. Another one is the abundance conjecture asserting that K_X is semi-ample when it is nef. In our Equivariant Minimal Model Program, we will overcome these two obstacles with the help of dynamics.

We end this section by citing two useful results on the two conjectures. Hashizume and Hu [27, Theorem 1.7] proved the special termination of directed flips for non-pseudo-effective pairs. This is the only termination needed in our EMMP.

Theorem 6 *Let (X, Δ) be a $\mathbb{Q}$-factorial log canonical pair over S with $K_X + \Delta$ not being pseudo-effective over S. Then there is a finite sequence of $(K_X + \Delta)$-MMP over S*

$$(X, \Delta) = (X_2, \Delta_1) \dashrightarrow (X_2, \Delta_2) \dashrightarrow \cdots \dashrightarrow (X_n, \Delta_n) \to Y$$

where $X_i \dashrightarrow X_{i+1}$ is birational and $(X_n, \Delta) \to Y$ is a Fano contraction.

Gongyo [25] proved the abundance conjecture for the numerically trivial log canonical pair.

Theorem 7 *Let (X, Δ) be a log canonical pair with $K_X + \Delta \equiv 0$. Then $K_X + \Delta \sim_{\mathbb{Q}} 0$.*

3 Various Types of Endomorphisms and Fundamental Properties

In this section, we study several important endomorphisms and their propositions, as a preparation for later sections.

We begin with the following (cf. [40, 53, 54, 58]):

Definition 8 Let $f : X \to X$ be a surjective endomorphism of a projective variety X.

1. f is *q-polarized* if $f^*L \sim qL$ for some ample Cartier divisor L and integer $q > 1$.
2. f is *amplified* (in the sense of [40], which is also the assumption of Theorem 32 in [15]) if $f^*L - L = H$ for some Cartier divisor L and ample divisor H.
3. f is *int-amplified* if $f^*L - L = H$ for some ample Cartier divisors L and H, i.e., the operator $(f^* - \mathrm{id})$ maps an **int**erior point of the nef cone $\mathrm{Nef}(X)$ to another **int**erior point of the same cone and is like an **int**ernal mapping in the sense of Theorem 10 (4), whence the naming of this notion.
4. f is *quasi-amplified* if $f^*L - L = H$ for some Cartier divisor L and big Cartier divisor H.
5. f is *PCD* (a short form for Periodic Countable Dense) if the set $\mathrm{Per}(f_K)$ of periodic points is countable and Zariski dense in X_K for some uncountable algebraically closed field extension K/k.
6. f is of *positive entropy* if the spectral radius of $f^*|_{\mathrm{N}^1(X)}$ on the Neron-Severi real space $\mathrm{N}^1(X) = \mathrm{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ is greater than 1 (see e.g. [14]).
7. f is of *null entropy* if f is not of positive entropy.
8. f is *p-cohomologically hyperbolic* if $\delta_p(f) > \delta_i(f)$ for any $i \in \{1, 2, \ldots, \dim X\} \backslash \{p\}$, where $\delta_i(f) = \lim_{n \to +\infty} ((f^n)^* H^i \cdot H^{\dim(X)-i})^{1/n}$ for some (and hence any) nef and big divisor H (cf. [13]).

By cone analysis and the Brauer fixed point theorem, a criterion for polarized endomorphisms is given by [56, Propositions 1.1 and 2.9].

Proposition 9 *Let $f : X \to X$ be a surjective endomorphism of a projective variety X and $q > 0$ a rational number. Assume one of the following conditions.*

1. *$f^*H \equiv qH$ for some big $\mathbb{R}$-Cartier divisor H.*
2. *X is normal and $f^*H \equiv qH$ for some big Weil $\mathbb{R}$-divisor H.*
3. *$f^*|_{\mathrm{NS}_{\mathbb{C}}(X)}$ is diagonalizable with all the eigenvalues being of modulus q.*

Then q is an integer and $f^*A \equiv qA$ for some ample Cartier divisor A. Further, if $q > 1$, then f is polarized.

A criterion for int-amplified endomorphisms is given in [53, Theorem 1.1]. We refer to [76, Theorem 1.1] for a similar criterion in the Kähler setting.

Theorem 10 (see also Proposition 14) *Let $f : X \to X$ be a surjective endomorphism of a projective variety X. Then the following are equivalent.*

(1) The endomorphism f is int-amplified.
(2) All the eigenvalues of $\varphi := f^*|_{\mathrm{N}^1(X)}$ are of modulus greater than 1.
(3) There exists some big $\mathbb{R}$-Cartier divisor B such that $f^*B - B$ is big.
(4) If C is a φ-invariant convex cone in $\mathrm{N}^1(X)$, then $\emptyset \neq (\varphi - \mathrm{id}_{\mathrm{N}^1(X)})^{-1}(C) \subseteq C$.

We have the following easy application.

Proposition 11 *Consider the equivariant dynamical systems*

$$f \circlearrowright X \overset{\pi}{\dashrightarrow} Y \circlearrowleft g$$

where π is dominant. Then the following hold.

1. *If f is q-polarized (resp. int-amplified), then so is g.*
2. *Suppose $\dim X = \dim Y$. Then f is q-polarized (resp. int-amplified) if g is.*

A criterion for amplified endomorphisms is given in [54, Proposition 3.1].

Proposition 12 *Let $f : X \to X$ be a surjective endomorphism of a projective variety X. Then the following are equivalent.*

1. *f is amplified.*
2. *For any $Z \in \overline{\mathrm{NE}}(X)$, $f_*Z \equiv Z$ implies $Z \equiv 0$.*

We give a proof to the well-known result below since no reference is available. This result is applicable to varieties of Fano type or Mori Dream Spaces.

Proposition 13 *Let $f : X \to X$ be an amplified (or quasi-amplified) endomorphism of a projective variety X. Suppose either the nef cone $\mathrm{Nef}(X)$ or the closed cone $\mathrm{PE}(X)$ of pseudo-effective divisors is a rational polyhedron. Then f is int-amplified.*

Proof We give a proof for $\mathrm{PE}(X)$. For $\mathrm{Nef}(X)$, the proof is similar. Since $\mathrm{PE}(X)$ is a rational polyhedron and spans $\mathrm{N}^1(X)$, our f^* fixes all the (finitely many) extremal rays of $\mathrm{PE}(X)$ after iteration and hence $f^*|_{\mathrm{N}^1(X)}$ is a diagonal action. Let $v_1, \ldots, v_n$ be a basis of $\mathrm{NS}_{\mathbb{Q}}(X)$ such that $\mathbb{R}_{\geq 0} v_i$ are all extremal rays of $\mathrm{PE}(X)$. Write $f^* v_i = r_i v_i$ with $r_i > 0$. Note that r_i is also an algebraic integer since it is an eigenvalue of $f^*|_{\mathrm{NS}(X)/(\mathrm{torsion})}$. So r_i is a positive integer. Assume that $r_i = 1$ for $i \leq t$ and $r_i > 1$ for $i > t$. By the assumption, $f^*D - D = A$ is big for some divisor D. Write $D = \sum_{i=1}^{n} a_i v_i$ in $\mathrm{N}^1(X)$. Note that $A = \sum_{i=t+1}^{n} a_i(r_i - 1)v_i$ is big. Hence $A' := \sum_{i=t+1}^{n} |a_i|(r_i - 1)v_i$ is big too, since v_i is pseudo-effective. Now $f^*A' - A' = \sum_{i=t+1}^{n} |a_i|(r_i - 1)^2 v_i$ is big too. Thus f is int-amplified by Theorem 10. □

Thanks to Matsuzawa for bringing the following nice equivalence to our attention.

Proposition 14 *Let $f : X \to X$ be a surjective endomorphism of a projective variety X of dimension n. Then f is int-amplified if and only if f is n-cohomologically hyperbolic.*

Proof By the log concavity of $\delta_k(f)$ (cf. [26, Theorem 1.2]), f is n-cohomologically hyperbolic if and only if $\delta_{n-1}(f) < \delta_n(f) = \deg f$. Note that $\delta_{n-1}(f)$ is the spectral radius of $f^*|_{\mathrm{N}_1(X)}$. By the projection formula, $f_* f^* = (\deg f)\,\mathrm{id}$ on $\mathrm{N}_1(X)$ and $f^*D \cdot C = D \cdot f_* C$ for any $D \in \mathrm{N}^1(X)$ and $C \in \mathrm{N}_1(X)$. So the smallest modulus α of eigenvalues of $f_*|_{\mathrm{N}_1(X)}$ is $\deg f/\delta_{n-1}(f) > 1$. Note that $f^*|_{\mathrm{N}^1(X)}$ and $f_*|_{\mathrm{N}_1(X)}$ have the same eigenvalues since they are dual to each other. Therefore, $\delta_{n-1}(f) < \deg f$ is equivalent to that all the eigenvalues of $f^*|_{\mathrm{N}^1(X)}$ have modulus > 1. Then we apply Theorem 10. □

The following results ([54, Theorems 1.5 and 1.7]) give characterizations and comparison of amplified and PCD endomorphisms of abelian varieties in terms of their actions on the cohomology groups and divisor classes.

Theorem 15 *Let $f : A \to A$ be a surjective endomorphism of an abelian variety A. Then the following hold.*

(1) f is amplified if and only if no eigenvalue of $f^*|_{H^1(A,\mathcal{O}_A)}$ is of modulus 1.
(2) f is PCD if and only if no eigenvalue of $f^*|_{H^1(A,\mathcal{O}_A)}$ is a root of unity.

Theorem 16 *Let $f : A \to A$ be a surjective endomorphism of an abelian variety A. Then the following hold.*

(1) f is amplified if and only if $f^*D \not\equiv D$ for any nef $\mathbb{R}$-Cartier divisor $D \not\equiv 0$.
(2) f is PCD if and only if $f^*D \not\equiv D$ for any nef Cartier divisor $D \not\equiv 0$.

Remark 17

1. It is possible that all the eigenvalues of $f^*|_{\mathrm{N}^1(X)}$ are q, but f is not polarized, or equivalently, $f^*|_{\mathrm{N}^1(X)}$ is not diagonalizable, see [53, Example 10.1].
2. Though polarized endomorphisms are always int-amplified, it is possible for a projective variety to admit an int-amplified endomorphism but no polarized endomorphism, see the example constructed in [50, Section 7].
3. Though the product of polarized endomorphisms is int-amplified, it is possible that an int-amplified endomorphism may never split as or even be dominated by a product of polarized endomorphisms, see [53, Example 10.3]. We also refer to [61] for a splitting result.
4. Some K3 surface admits an automorphism of positive entropy that is not PCD, and some K3 surface admits a PCD automorphism that is not amplified, see [54, Examples 5.7 and 5.8].

We end this section with the relation between int-amplified endomorphism and Bott vanishing, due to Kawakami and Totaro [37, Theorem C]. This way, they gave a different proof (without using EMMP) of Fano threefolds being toric when admitting int-amplified endomorphisms, see Remark 42. Recall that the sheaf of reflexive differentials $\Omega_X^{[j]}$ is the double dual $(\Omega_X^j)^{**}$.

Theorem 18 *Let X be a normal projective variety over a perfect field k. Suppose that X admits an int-amplified endomorphism whose degree is invertible in k. Then X satisfies Bott vanishing for ample Weil divisors. That is,*

$$H^i(X, \Omega_X^{[j]}(A)) = 0$$

for every $i > 0$, $j \geq 0$, and any ample ($\mathbb{Q}$-Cartier) Weil divisor A.

4 Int-Amplified Equivariant Minimal Model Program (EMMP)

In this section, we develop the Equivariant Minimal Model Program (EMMP) for arbitrary surjective endomorphisms of a $\mathbb{Q}$-factorial klt projective variety X that admits an int-amplified endomorphism. This way, one is reduced to the study of endomorphisms on Q-abelian varieties and Fano varieties. Here, Q-abelian variety means a quasi-étale (étale in codimension 1) quotient of an abelian variety and Fano variety means the anti-canonical divisor $-K_X$ is ample.

Let $\mathrm{SEnd}(X)$ be the set of *all surjective endomorphisms* on X. A submonoid G of a monoid Γ is said to be of *finite-index* in Γ if there is a chain $G = G_0 \leq G_1 \leq \cdots \leq G_r = \Gamma$ of submonoids and homomorphisms $\rho_i : G_i \to F_i$ such that $\mathrm{Ker}(\rho_i) = G_{i-1}$ and all the F_i are finite *groups*.

An lc pair (X, Δ) may have infinitely many $(K_X + \Delta)$-negative extremal rays. However, this case will never happen if X further admits an int-amplified endomorphism; moreover, every MMP starting from X is equivariant relative to $\mathrm{SEnd}(X)$,

up to finite-index (cf. [57, Theorem 1.1]). The key idea of the proof is based on the finiteness of totally periodic subvarieties and the ramification divisor formula.

Theorem 19 *Let X be a normal projective variety admitting an int-amplified endomorphism. Then we have:*

(1) Suppose (X, Δ) is lc for some effective $\mathbb{R}$-divisor Δ. Then there are only finitely many $(K_X + \Delta)$-negative extremal rays.
(2) Suppose X is $\mathbb{Q}$-factorial. Then any finite sequence of MMP starting from X is G-equivariant for some finite-index submonoid G of $\mathrm{SEnd}(X)$.

The detailed EMMP ([57, Theorem 1.2]) can be further described. The result of Y being Q-abelian requires techniques on Galois closure and vanishing of the first and second Chern classes.

Theorem 20 *Let $f : X \to X$ be an int-amplified endomorphism of a $\mathbb{Q}$-factorial klt projective variety X. Then there exist a finite-index submonoid G of $\mathrm{SEnd}(X)$, a Q-abelian variety Y, and a G-EMMP over Y*

$$X = X_0 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots \dashrightarrow X_r = Y$$

(i.e. $\forall\ g \in G = G_0$ descends to $g_i \in G_i$ on each X_i), with every $X_i \dashrightarrow X_{i+1}$ a divisorial contraction, a flip or a Fano contraction, of a K_{X_i}-negative extremal ray, such that:

(1) There is a finite quasi-étale Galois cover $A \to Y$ from an abelian variety A such that $G_Y := G_r$ lifts to a submonoid G_A of $\mathrm{SEnd}(A) \leq \mathrm{End}_{\mathrm{variety}}(A)$.
(2) If g in G is polarized (resp. int-amplified) then so are its descending g_i on X_i and the lifting to A of g_r on $X_r = Y$.
(3) If g in G is amplified and its descending g_r on X_r is int-amplified, then g is also int-amplified.
(4) If K_X is pseudo-effective, then $X = Y$ and it is Q-abelian.
(5) If K_X is not pseudo-effective, then for each i, $X_i \to Y$ is equi-dimensional well-defined with every fibre (irreducible) rationally connected (in the sense of [38, IV, Definition 3.2]). The $X_{r-1} \to X_r = Y$ is a Fano contraction.
(6) For any subset $H \subseteq G$ and its descending $H_Y \leq \mathrm{SEnd}(Y)$, and any field K of characteristic 0, H acts on $\mathrm{NS}_K(X)$ as commutative diagonal matrices with respect to a suitable basis if and only if so does H_Y.

Yoshikawa [73, Theorem 1.3] recently constructed, in a sophisticated way, a new sequence of Equivariant log MMP (called "maximal sequence of steps of MMP of canonical bundle formula type"), together with a new terminology "f-pair", providing a new perspective on the ramification divisor. This way, he showed the following:

Theorem 21 *Let $f : X \to X$ be an int-amplified endomorphism of a $\mathbb{Q}$-factorial klt projective variety. Then there exists an f-equivariant quasi-étale cover $\mu : \widetilde{X} \to X$ such that the albanese morphism $\mathrm{alb} : \widetilde{X} \to A$ is a fiber space and $\widetilde{X}$ is of Fano*

type over A, i.e., (X, Δ) is klt and $-(K_X + \Delta)$ is ample over A for some effective $\mathbb{Q}$-divisor Δ.

Remark 22 We refer to [53, 56, 57, 66, 74] for historical notes. We also refer to [11, 58, 76] for int-amplified EMMP in the setting of positive characteristics or Kähler spaces.

5 Non-isomorphic EMMP for Projective Surfaces

In this section, we study normal projective surfaces admitting a non-isomorphic surjective endomorphism.

Nakayama [64] first showed the finiteness of negative curves on a smooth projective surface admitting a non-isomorphic surjective endomorphism. This result can be further generalised to contractible curves on normal projective surfaces, see [59, Lemma 4.3]. Below is a beautiful result of Wahl [70, Theorem 2.8], which has been further generalized to higher dimensional cases, see Sect. 9.

Theorem 23 *Let X be a normal projective surface admitting a non-isomorphic surjective endomorphism. Then X has at worst log canonical singularities. In particular, the canonical divisor K_X is $\mathbb{Q}$-Cartier.*

Using the above result, we can show the following (see [59, Theorem 4.7]).

Theorem 24 *Let X be a normal projective surface admitting a non-isomorphic surjective endomorphism. Then any MMP starting from X is G-equivariant for some finite-index submonoid G of* $\mathrm{SEnd}(X)$.

Based on the above theorem, the structural result ([32, Theorem 1.1]) can be given. We refer to [20] for the Minimal Model Program of log canonical projective surfaces.

Theorem 25 *Let $f : X \to X$ be a non-isomorphic surjective endomorphism of a normal projective surface. Then X has only log canonical (lc) singularities. If the canonical divisor K_X is pseudo-effective, then Case (1) below occurs. If K_X is not pseudo-effective, replacing f by an iteration, we may run an f-Equivariant Minimal Model Program (EMMP)*

$$X = X_1 \longrightarrow \cdots \longrightarrow X_j \longrightarrow \cdots \longrightarrow X_r \longrightarrow Y,$$

contracting K_{X_j}-negative extremal rays, with $X_j \to X_{j+1}$ $(j < r)$ being divisorial and $X_r \to Y$ being Fano contraction, such that one of the following cases occurs.

1. *f is quasi-étale, i.e., étale in codimension* 1*; there exists an f-equivariant quasi-étale finite Galois cover $\nu : V \to X$ with V smooth as in [32, Theorem 2.14] and the lifting $f|_V$ of f is étale.*

2. *Y is a smooth projective curve of genus $g(Y) \geq 1$; f descends to an automorphism of finite order on the curve Y.*
3. *Y is an elliptic curve; $X \to Y$ is a $\mathbb{P}^1$-bundle.*
4. *$Y \cong \mathbb{P}^1$; there is an $f|_{X_r}$-equivariant finite surjective morphism $X_r \to Y \times \mathbb{P}^1$.*
5. *$Y \cong \mathbb{P}^1$; f is polarized; $K_X + S \sim_{\mathbb{Q}} 0$ for an f^{-1}-stable reduced divisor S; there is an f-equivariant quasi-étale finite Galois cover $\nu : V \to X$ where V is a ruled surface over an elliptic curve and $\nu^* S$ is a disjoint union of two cross-sections (and the lifting of f is étale outside $\nu^* S$).*
6. *$Y \cong \mathbb{P}^1$; f is polarized; there exists an equivariant commutative diagram:*

$$\begin{array}{ccc} \widetilde{f} \circlearrowright \widetilde{X} & \xrightarrow{\mu_X} & X \circlearrowleft f \\ \downarrow \widetilde{\pi} & & \downarrow \pi \\ g_E \circlearrowright E & \longrightarrow & Y \circlearrowleft g \end{array}$$

   here E is an elliptic curve; $\widetilde{X}$ is the normalisation of $X \times_Y E$; $\widetilde{\pi}$ is a $\mathbb{P}^1$-bundle; π is a $\mathbb{P}^1$-fibration; $\widetilde{f}$ and g_E are finite surjective endomorphisms; μ_X is quasi-étale.
7. *$Y \cong \mathbb{P}^1$; X_r is of Fano type; there is an $f|_{X_r}$-equivariant birational morphism $X_r \to \overline{X}$ to a klt Fano surface with Picard number $\rho(\overline{X}) = 1$; every X_j $(1 \leq j \leq r)$ is a rational surface whose singularities are klt (hence $\mathbb{Q}$-factorial).*
8. *Y is a point; X_r is a projective cone over an elliptic curve E; the normalisation Γ of the graph of $X \dashrightarrow E$ is a $\mathbb{P}^1$-bundle, and f lifts to $f|_\Gamma$ such that $(f|_\Gamma)^*|_{\mathrm{N}^1(\Gamma)} = \delta_f \,\mathrm{id}$.*
9. *Y is a point and hence $\rho(X_r) = 1$; $-K_{X_r}$ is ample; every X_j $(1 \leq j \leq r)$ is a rational surface whose singularities are lc and rational (hence $\mathbb{Q}$-factorial).*

Remark 26 We refer to [64] for the precise list of birationally ruled surfaces which admit non-isomorphic surjective endomorphisms.

6 Étale EMMP for Smooth Projective Threefolds

In this section, we study non-isomorphic étale endomorphisms of smooth projective threefolds.

Let $f : X \to X$ be a surjective endomorphism of a normal projective variety. We have the ramification divisor formula

$$K_X = f^* K_X + R_f$$

where R_f is the ramification divisor. Recall that f is quasi-étale or étale in codimension one if $R_f = 0$. Note that K_X being pseudo-effective implies that f is quasi-étale. If X is smooth and f is quasi-étale, then f is further étale by the purity of branch loci.

When the Kodaira dimension $\kappa(X) > 0$, we can apply the equivariant Iitaka fibration ([65, Theorem A]).

Theorem 27 *Let $f : X \dashrightarrow X$ be a dominant self-map of a smooth projective variety (or a compact Kähler manifold) X. Then the Iitaka fibration $\phi : X \dashrightarrow Y$ is f-equivariant and $f|_Y$ is an automorphism of finite order.*

In the following, we focus on a smooth projective threefold X.

Suppose $\kappa(X) \geq 0$. This is equivalent to K_X being pseudo-effective by the abundance for threefolds (cf. [39, Remark 3.13]). Moreover, X has no Fano contraction and the first birational contraction, which exists when K_X is not nef, has to be a divisorial contraction (cf. [62, Theorem 3.3]). In particular, the closed cone $\overline{\mathrm{NE}}(X)$ of effective 1-cycles has only finitely many K_X-negative extremal rays (cf. [17, Proposition 4.6]). With the help of non-isomorphic étale dynamics, Fujimoto [17, Theorem 4.8] further showed that the divisorial contraction has to be (the inverse of) the blowup along an elliptic curve. In particular, one can run the following smooth EMMP.

Theorem 28 *Let $f : X \to X$ be a non-isomorphic étale endomorphism of a smooth projective threefold X with $\kappa(X) \geq 0$. Then after iteration of f, any MMP starting from X is f-equivariant and involves only divisorial contractions which are (the inverse of) the blowups along elliptic curves.*

Furthermore, Nakayama and Fujimoto [19, Main Theorem] completely classified smooth projective threefolds which have non-negative Kodaira dimension and admit non-isomorphic surjective endomorphisms.

Theorem 29 *Let X be a smooth projective threefold with pseudo-effective K_X. Then the following conditions are equivalent to each other:*

(A) X admits a non-isomorphic surjective endomorphism.
(B) There exist a finite étale Galois covering $\tau : \widetilde{X} \to X$ and an abelian scheme structure $\varphi : \widetilde{X} \to T$ over a variety T of dimension ≤ 2 such that the Galois group $\mathrm{Gal}(\tau)$ acts on T and φ is $\mathrm{Gal}(\tau)$-equivariant.

Now we consider the case when the Kodaira dimension $\kappa(X) < 0$, or equivalently, K_X is not pseudo-effective. Of course, we can run MMP which ends with a Mori fibre space. However, the MMP is no longer equivariant in general. Here is an example.

Example 30 There exists a smooth rational surface S admitting an automorphism g of positive entropy with no g-periodic curves (cf. [42]). Then we cannot run any g-EMMP. Let E be an elliptic curve and n_E ($n \geq 2$) the multiplication map. Take $X = S \times E$ and $f = g \times n_E$. Clearly, $\kappa(X) < 0$ and f is étale. Note that X admits no f^{-1}-periodic prime divisor. Indeed, let P be an f^{-1}-invariant prime divisor and $e \in E$ the identity element. Then $P \cap S \times \{e\}$ is $(f|_{S\times\{e\}})^{-1}$-invariant and hence $P = S \times \{a\}$ for some a. However, E admits no $(n_E)^{-1}$-periodic point. In particular, we cannot run any f-EMMP while S and hence X are clearly not relatively minimal.

Nevertheless, Fujimoto established the étale sequence of constant Picard number by running a not necessarily "Equivariant" (even after iteration) MMP. We refer to [18] and its sequels for the details.

7 Quasi-Amplified Endomorphisms of Klt Log Calabi-Yau Varieties

In this section, we shall further study quasi-amplified endomorphisms; see Definition 8.

A normal projective variety X is of *Calabi-Yau type*, if (X, Δ) is log canonical and $K_X + \Delta \equiv 0$ for some effective Weil $\mathbb{Q}$-divisor Δ. The pair is called log Calabi-Yau. Such a pair has $K_X + \Delta \sim_{\mathbb{Q}} 0$ ($\mathbb{Q}$-linear equivalence) by Theorem 7. If the pair is further klt, we say X is of klt Calabi-Yau type.

One can run the equivariant contractions for quasi-amplified endomorphism on X of klt Calabi-Yau type as proved in [54].

Theorem 31 *Let $f : X \to X$ be a quasi-amplified endomorphism of a projective variety X of klt Calabi-Yau type. Then replacing f by a positive power, there is an f-equivariant sequence of birational contractions of extremal rays (in the sense of [57, Definition 4.1])*

$$X = X_1 \to \cdots \to X_i \to \cdots \to X_r$$

(i.e. $f = f_1$ descends to surjective endomorphism f_i on each X_i), such that we have:

(1) f_r is amplified.
(2) For each i, X_i is of klt Calabi-Yau type.
(3) For each i, f_i is of positive entropy and quasi-amplified and $\mathrm{Per}(f_i) \cap U_i$ is countable and Zariski dense in X_i for some open subset $U_i \subseteq X_i$.
(4) Suppose the base field k is uncountable. For each i, f_i has a Zariski dense orbit.

The density of periodic points $\mathrm{Per}(f_i)$ is based on a famous result of Fakhruddin [15].

Theorem 32 *Let $f : X \to X$ be an amplified endomorphism of a projective variety over an algebraically closed field k of arbitrary characteristic. Then the set $\mathrm{Per}(f)$ of periodic points is Zariski dense in X.*

As an application of Theorem 31 to Hyperkähler manifolds, we have:

Theorem 33 *Let $f : X \to X$ be an automorphism of a projective Hyperkähler manifold X. Then the following are equivalent.*

(1) f is of positive entropy.
(2) $f^*D \not\equiv D$ for any nef $\mathbb{R}$-Cartier divisor $D \not\equiv 0$.
(3) f is quasi-amplified.
(4) For some $n > 0$, f^n is birationally equivalent to some amplified automorphism $f' : X' \to X'$.

Moreover, if f is PCD, then all the above are satisfied.

8 Fano Threefolds with Non-isomorphic Endomorphisms

In this section, we characterize smooth Fano threefolds admitting non-isomorphic surjective endomorphisms ([60, Main Theorem, Theorems 1.3 and 1.4]).

Theorem 34 *A smooth Fano threefold is either toric or a product of* $\mathbb{P}^1$ *and a del Pezzo surface if and only if it admits a non-isomorphic surjective endomorphism.*

To be precise, we have the following characterization. Note that a surjective endomorphism f on a Fano variety or more generally on a Mori Dream Space, is amplified if and only if it is int-amplified (cf. Proposition 13).

Theorem 35 *Let X be a smooth Fano threefold. Then the following are equivalent.*

(1) X is a product of $\mathbb{P}^1$ and a del Pezzo surface.
(2) X admits a non-isomorphic surjective endomorphism which is not polarized even after iteration.
(3) X admits a non-isomorphic surjective endomorphism which is non-amplified (or equivalently, non-int-amplified).

Theorem 36 *Let X be a smooth Fano threefold. Then the following are equivalent.*

(1) X is a toric variety.
(2) X admits a polarized endomorphism.
(3) X admits an amplified (or equivalently, int-amplified) endomorphism.

Theorem 36 gives a partial answer to the following question, originally proposed by Fakhruddin [15, Question 4.4] for polarized endomorphisms and later generalised to int-amplified cases ([61, Question 1.2]). Question 37 itself generalizes a long-standing folklore Conjecture of the 1980s, which focuses on Fano manifolds of Picard number 1.

Question 37 Let X be a rationally connected smooth projective variety. Suppose that X admits an int-amplified (or polarized) endomorphism f. Is X a toric variety?

Now we briefly explain the strategy of the proofs of the theorems in this section.

Recall that a smooth projective variety is *Fano* if $-K_X$ is ample. By [8, Corollary 1.3.2], a Fano manifold is a Mori Dream Space. So $\overline{\mathrm{NE}}(X)$ is a rational polyhedron (this can also be obtained by the cone theorem). Then we can run EMMP for any surjective endomorphism $f : X \to X$.

The proof starts with the following Picard number 1 case, due to Amerik, Rovinsky and Van de Ven [1, 3], see also [30] for a different approach.

Theorem 38 *Let X be a smooth Fano threefold of Picard number* 1 *admitting a non-isomorphic surjective endomorphism. Then $X \cong \mathbb{P}^3$.*

The Picard number ≤ 2 case was settled in [60, Theorem 5.1].

Theorem 39 *Let X be a smooth Fano threefold of Picard number $\rho(X) \leq 2$ admitting a non-isomorphic surjective endomorphism f. Then X is toric. To be precise, X is either $\mathbb{P}^3$, or a splitting $\mathbb{P}^1$-bundle over $\mathbb{P}^2$, or a blowup of $\mathbb{P}^3$ along a line.*

For the case of Picard number ≥ 3, we obtain a conic bundle structure ([60, Theorem 6.1]). Here a fibration $\tau : X \to Y$ of smooth projective varieties is a *conic bundle* if every fibre is isomorphic to a conic, i.e., a scheme of zeros of a nonzero homogeneous form of degree 2 on $\mathbb{P}^2$. Denote by $\Delta_\tau := \{y \in Y \mid \tau \text{ is not smooth over } y\}$ the *discriminant* locus of τ. When Y is a surface, either $\Delta_\tau = \emptyset$ or Δ_τ is a (not necessarily irreducible) curve with only ordinary double points (cf. [63, Sect. 6]). If X is further assumed to be Fano, then we say τ is a *Fano conic bundle*.

Theorem 40 *Let X be a smooth Fano threefold with $\rho(X) \geq 3$. Suppose that X admits a non-isomorphic surjective endomorphism f. Then X admits a (Fano) conic bundle $\tau : X \to Y$.*

With the help of the Fano conic bundle structure, we can run the relative Minimal Model Program as discussed in [63, Sect. 6] and obtain the following special EMMP in the category of smooth Fano threefolds as stated in [60, Theorem 6.2].

Theorem 41 *Let X be a smooth Fano threefold with a conic bundle $\tau : X \to Y$. Suppose that X admits a non-isomorphic surjective endomorphism f. Then there exists an f-Equivariant (after iteration of f) Minimal Model Program (EMMP)*

$$X = X_1 \to \cdots \to X_i \to \cdots \to X_{r+1} \to Y$$

such that the following hold.

1. *$r = \rho(X) - \rho(Y) - 1$ and each X_i is a smooth Fano threefold.*
2. *$\tau_{r+1} : X_{r+1} \to Y$ is a Fano contraction and an algebraic $\mathbb{P}^1$-bundle $\mathbb{P}_Y(\mathcal{E})$ over Y.*
3. *The composition $\tau_i : X_i \to Y$ is a conic bundle with $(f|_Y)^{-1}$-invariant discriminant $\Delta_{\tau_i} = C_i \cup \cdots \cup C_r$ a disjoint union of $r + 1 - i$ smooth curves on Y.*
4. *The composition $\pi : X \to X_{r+1}$ is the blowup of X_{r+1} along r disjoint union of $(f|_{X_{r+1}})^{-1}$-invariant smooth curves $\bigcup_{i=1}^{r} \overline{C_i}$ with $\tau_{r+1}(\overline{C_i}) = C_i$.*

There remain two difficult issues:

- Confirming the splitting of $\mathcal{E}$ when $X = \mathbb{P}_Y(\mathcal{E})$ for some rank 2 vector bundle $\mathcal{E}$ on Y, and
- Verifying that each blowup in the EMMP is a toric blowup.

The splitting problem was dealt with in [60, Theorems 6.4 and 7.1] for the respectively polarized and non-polarized cases. However, the EMMP in general is not a sequence of toric blowups. Indeed, one needs to choose very carefully a suitable EMMP to reach this goal as discussed in the proof of [60, Theorems 8.1 and 7.1] for the respectively int-amplified and non-int-amplified cases.

Remark 42

1. Theorem 36 gives a partial answer (only for smooth Fano threefolds) to a question proposed by Fakhruddin [15, Question 4.4].

2. Jia and Zhong further developed the technique and proved that a smooth Fano fourfold admitting a conic bundle structure and an int-amplified endomorphism is indeed toric, see [33].
3. Totaro classified smooth Fano 3-folds satisfying the Bott vanishing ([69, Theorem 0.1]). Kawakami and Totaro proved that every normal projective variety admitting an int-amplified endomorphism f satisfies the Bott vanishing (cf. Theorem 18). This way, Totaro [68] obtained a different proof of Theorem 36, which works for the known list of smooth Fano threefolds defined over an algebraically closed field k with $\deg f$ invertible in k.

9 Application to Broustet-Gongyo Conjecture

A generalisation of Question 37 is the following conjecture proposed by Broustet and Gongyo [9]. They confirmed the conjecture for surfaces and Mori Dream Spaces, using the EMMP. Recall that a normal projective variety X is of *Calabi-Yau type* if (X, Δ) is an lc pair for some effective Weil $\mathbb{Q}$-divisor Δ such that $K_X + \Delta \sim_{\mathbb{Q}} 0$. By the abundance (cf. Theorem 7), the latter condition is equivalent to $K_X + \Delta \equiv 0$. We also call such a pair (X, Δ) *log Calabi-Yau*.

Conjecture 43 Let X be a normal projective variety admitting a polarized endomorphism. Then X is of Calabi-Yau type.

For the X in the above conjecture, assuming further that the canonical divisor K_X is $\mathbb{Q}$-Cartier, Broustet and Höring [10] proved that X has at worst log canonical singularities; see also Theorem 23. Conjecture 43 has been verified for rationally connected smooth projective varieties by Yoshikawa since Fano type is Calabi-Yau type; see [73] or Theorem 21. Based on this, the following theorem in [55] fully solved the case of smooth projective threefolds.

Theorem 44 *Let X be a smooth projective threefold admitting a polarized endomorphism f. Then X is of Calabi-Yau type.*

We briefly explain the strategy of the proof of Theorem 44. We first run the f-Equivariant Minimal Model Program which ends with a Q-abelian variety Y, i.e., a quasi-étale (or étale in codimension one) quotient of an abelian variety, see Theorem 20. Since the smooth rationally connected case has been verified by Yoshikawa, one may assume $\dim Y > 0$. Then we study the fibration $\pi : X \to Y$ and its f-periodic general fibres. By applying the canonical bundle formula and ramification divisor formula, the natural idea is to reduce the problem to the following Conjecture 45 (for surfaces) proposed by Gongyo.

Conjecture 45 (Gongyo) Let $f : X \to X$ be a q-polarized endomorphism of a smooth projective variety X. Then $(X, \frac{R_f}{q-1})$ is an lc pair after iteration of f.

However, in dimension two, Conjecture 45 is not known even for $\mathbb{P}^2$, which is, in fact, the only leftover case. So the main difficulty of proving Theorem 44 remains

in the case where $\pi : X \to Y$ is a $\mathbb{P}^2$-bundle over an elliptic curve Y. By applying the Iitaka fibration of the anti-canonical divisor, we are reduced to the case where the anti-canonical divisor $-K_X$ is big (i.e., it is the sum of an ample divisor and an effective divisor), see [55, Sect. 5]. Then the problem boils down to the very concrete case: $X \cong \mathbb{P}_Y(\mathcal{F}_2 \oplus \mathcal{L})$ where $\mathcal{F}_2$ is the unique indecomposable rank 2 vector bundle of degree 0 with non-trivial global sections and $\mathcal{L}$ is a line bundle of negative degree.

10 Application to Kawaguchi-Silverman Conjecture

In this section, we work over a number field K with a fixed algebraic closure $\overline{K}$. Let $f : X \to X$ be a surjective endomorphism of a normal projective variety X. One may consider two dynamical invariants: the (first) *dynamical degree*

$$\delta_f = \lim_{n \to +\infty} ((f^n)^* H \cdot H^{\dim(X)-1})^{1/n}$$

of f and the *arithmetic degree*

$$\alpha_f(x) = \lim_{n \to +\infty} h_H(f^n(x))^{1/n}$$

of f at $x \in X(\overline{K})$. Here H is any ample divisor on X and h_H is (the maximum of 1 and) the Weil height function associated with H. We refer to [29, Part B], [36], or [44, Sect. 2.2] for the detailed definition of the *Weil height function*. We refer to [46, Sect. 2] for some basic properties of these invariants.

The dynamical degree reflects the geometric complexity of iterations of f. On the other hand, the arithmetic degree reflects the arithmetic complexity of the given (forward) *f-orbit* $O_f(x) := \{x, f(x), f^2(x), \ldots\}$.

One naturally likes to compare these two invariants. By Kawaguchi–Silverman [35], the arithmetic degree at any point is less than or equal to the dynamical degree (see also [44]). Then, one wonders when the arithmetic degree at a point attains (its upper bound) the dynamical degree. Kawaguchi and Silverman proposed (see [35]):

Conjecture 46 (*Kawaguchi-Silverman Conjecture, KSC for short*) Let $f : X \to X$ be a surjective endomorphism of a projective variety X defined over a number field K. Let $x \in X(\overline{K})$ such that $\alpha_f(x) < \delta_f$. Then the orbit $O_f(x)$ is not Zariski dense in $X_{\overline{K}} := X \times_K \overline{K}$.

Remark 47

1. The original conjecture in [35] has also been formulated for rational maps. For endomorphisms, Kawaguchi and Silverman proved that $\alpha_f(x)$ equals either 1 or the modulus of an eigenvalue of $f^*|_{\mathrm{NS}(X)}$. For rational maps, the existence of the limit defining $\alpha_f(x)$ is the first part of the original conjecture, and Matsuzawa [44, Theorem 1.4] proved that the supremum limit still satisfies

$$\overline{\alpha}_f(x) := \limsup_{n\to+\infty} h_H(f^n(x))^{1/n} \leq \delta_f.$$

2. The KSC was further generalised to the (stronger) small Arithmetic Non-Density conjecture (sAND for short) in [46], see [46, Conjecture 1.3 and Remark 1.4]. The sAND conjecture implies the KSC and is much harder than the KSC with obstacles proposed as three questions in [46, Sect. 8]. We refer to [6, 47] for recent progress towards these questions.

We now summarise the known results on KSC till the very recent, to the best of our knowledge.

Theorem 48 *The KSC holds true for a surjective endomorphism f on a projective variety X which fits one of the following cases.*

1. *f is polarized (cf. [34]).*
2. *X is a Mori dream space (cf. [45]).*
3. *$\dim(X) \leq 2$ (for smooth case, see [48]; for singular case, see [59]).*
4. *X is a Hyperkähler manifold (cf. [43]).*
5. *X is a (Q)-abelian variety (cf. [67]).*
6. *X is a smooth rationally connected projective variety admitting an int-amplified endomorphism (see [59] and [51] for respectively threefolds and n-folds).*
7. *X is a projective threefold which admits an int-amplified endomorphism and has at worst $\mathbb{Q}$-factorial terminal singularities (cf. [46, Corollary 6.19]).*
8. *X is a smooth projective threefold with $q(X) > 0$ and f is an automorphism (cf. [12]).*

Remark 49 In fact, with ([46, Theorems 1.5 and 3.1]) we confirm the stronger sAND ([46, Conjecture 1.3]) and hence the KSC for Cases (1)–(7) in Theorem 48, see [46, Remark 1.4]. However, the sAND remains unsolved for Case (8) in Theorem 48.

Remark 50 When f is only a dominant self-map, KSC has been verified at least for the following cases.

1. f is certain self-map of $\mathbb{P}^n$ (cf. [34, Theorem 2]).
2. f is a non-isomorphic finite surjective endomorphism of a smooth affine surface of non-negative Kodaira dimension (cf. [31]).
3. f is birational and X is a smooth projective variety with Kodaira dimension 0 and irregularity $q(X) \geq \dim X - 1$ (cf. [12]).
4. X is a smooth projective variety with Kodaira dimension 0 and $q(X) = \dim X$ (cf. [12]).
5. A slightly weaker version of KSC (for generic orbits) is proved when f is 1-cohomologically hyperbolic, i.e., the first dynamical degree is larger than all the other dynamical degrees (cf. [49]).

We now briefly explain the EMMP used in proving Theorem 48 (3), (6) and (7).

When there is an int-amplified endomorphism, we are able to reduce KSC to the highly geometrically restrictive Case TIR by the following theorem (cf. [59, Theorem 1.7]). The reduction is based on the effectivity of $-K_X$ (cf. [59, Theorem 1.5]), the equivariant Chow reduction (cf. [59, Proposition 1.6]), and a carefully chosen EMMP (cf. [59, Propostion 9.2]).

Theorem 51 *Let X be a $\mathbb{Q}$-factorial klt projective variety admitting an int-amplified endomorphism. Then we have:*

(1) If K_X is pseudo-effective, then KSC holds for any surjective endomorphism of X.
(2) Suppose that KSC holds for Case TIR (for those $f|_{X_i} : X_i \to X_i$ appearing in any EMMP starting from X) to be defined below. Then KSC holds for any (not necessarily int-amplified) surjective endomorphism f of X.

Case TIR$_n$ (Totally Invariant Ramification case). Let X be a normal projective variety of dimension $n \geq 1$, which has only $\mathbb{Q}$-factorial Kawamata log terminal (klt) singularities and admits one int-amplified endomorphism. Let $f : X \to X$ be an arbitrary surjective endomorphism. Moreover, we impose the following conditions.

(A1) The anti-Kodaira dimension $\kappa(X, -K_X) = 0$; $-K_X$ is nef, whose class is extremal in both the *nef cone* $\mathrm{Nef}(X)$ and the *pseudo-effective divisors cone* $\mathrm{PE}^1(X)$.
(A2) $f^*D = \delta_f D$ for some prime divisor $D \sim_{\mathbb{Q}} -K_X$.
(A3) The ramification divisor of f satisfies $\mathrm{Supp}\ R_f = D$.
(A4) There is an f-equivariant Fano contraction $\pi : X \to Y$ with $\delta_f > \delta_{f|_Y}$ (≥ 1).

Indeed, we conjecture that Case TIR will never happen, which is confirmed for rationally connected smooth varieties (see [59] and [51] for threefolds and n-folds respectively) and is further evidenced by [46, Theorem 6.6], i.e., the following theorem, while the latter in turn, implies Theorem 48 (3), (6), and (7). For the surface case, we may remove the assumption on singularities in Case TIR (cf. [59, Theorem 5.2]), see also Theorem 25.

Theorem 52 *Suppose that the X in Case TIR has only ($\mathbb{Q}$-factorial) terminal singularities. Then* $\dim(X) \geq \dim(Y) + 3 \geq 4$.

For Theorem 48 (8), the essential case is when the Kodaira dimension $\kappa(X) = -\infty$, our f is an automorphism of positive entropy, and the Albanese map $\mathrm{alb} : X \to B$ is a surjective morphism to an elliptic curve B. Chen, Lin and Oguiso applied the classification result of Lesieutre [41, Theorem 1.7] to run smooth EMMP. This way, one can further narrow down to the case where X admits an f-equivariant conic bundle structure with the first dynamical degree preserved, thus reducing to the surface case, where KSC is known, see Theorem 48 (3).

11 Application to Zariski Dense Orbit Conjecture

In this section, we work over an algebraically closed field $\mathbf{k}$ of characteristic 0. We consider the following conjecture. Note that Conjecture 53(2) below is invariant under birational conjugation which is stronger than the long-standing Conjecture 53(1), called the Zariski dense orbit conjecture. In fact, they are equivalent modulo the Dynamical Mordell-Lang Conjecture, see [72, Proposition 2.6]. Recall that a rational function $\psi \in \mathbf{k}(X)$ is *f-invariant* if $f^*(\psi) := \psi \circ f = \psi$. Denote by $\mathbf{k}(X)^f$ the field of f-invariant rational functions on X. We have $\mathbf{k} \subseteq \mathbf{k}(X)^f \subseteq \mathbf{k}(X)$. Denote by $X(\mathbf{k})_f$ the set of points whose f-orbit $O_f(x)$ is well-defined.

Conjecture 53 Let X be an (irreducible) projective variety over $\mathbf{k}$ and $f : X \dashrightarrow X$ a dominant rational self-map such that $\mathbf{k}(X)^f = \mathbf{k}$. Then:

1. There is a point $x \in X(\mathbf{k})_f$ such that its f-orbit is Zariski-dense in X.
2. For every Zariski-dense open subset U of X, there exists a point $x \in X(\mathbf{k})_f$ whose orbit $O_f(x)$ is contained in U and Zariski-dense in X. In particular, the set

$$\{x \in X(k)_f : O_f(x) \text{ is Zariski-dense in } X\}$$

is Zariski dense in X.

We can extend Conjecture 53 (Zariski-topology version) to a stronger Conjecture 54 below (adelic-topology version). The adelic topology on $X(\mathbf{k})$ was proposed by Xie [72, §3] with the following properties.

1. It is stronger than the Zariski topology.
2. It is T_1, i.e., for any two distinct points $x, y \in X(\mathbf{k})$ there are adelic open subsets U, V of $X(\mathbf{k})$ such that $x \in U$, $y \notin U$ and $y \in V$, $x \notin V$.
3. Morphisms between algebraic varieties over $\mathbf{k}$ are continuous for the adelic topology.
4. Flat morphisms are open with respect to the adelic topology.
5. The irreducible components of $X(\mathbf{k})$ in the Zariski topology are the irreducible components of $X(\mathbf{k})$ in the adelic topology.
6. Let K be any subfield of $\mathbf{k}$ which is finitely generated over $\mathbb{Q}$ and such that X is defined over K and $\overline{K} = \mathbf{k}$. Then the action

$$\mathrm{Gal}(\mathbf{k}/K) \times X(\mathbf{k}) \to X(\mathbf{k})$$

is continuous with respect to the adelic topology.

When X is irreducible, Conjecture 53(5) implies that the intersection of finitely many non-empty adelic open subsets of $X(\mathbf{k})$ is non-empty. So, if $\dim X \geq 1$, the adelic topology is not Hausdorff. In general, the adelic topology is strictly stronger than the Zariski topology.

The adelic version of the Zariski-dense orbit conjecture was proposed in [72].

Conjecture 54 Assume that the transcendence degree of $\mathbf{k}$ over $\mathbb{Q}$ is finite. Let X be an irreducible variety over $\mathbf{k}$ and $f : X \dashrightarrow X$ a dominant rational map. If $\mathbf{k}(X)^f = \mathbf{k}$, then there exists a non-empty adelic open subset $A \subseteq X(\mathbf{k})$ such that the orbit of every point $x \in A \cap X(\mathbf{k})_f$ is Zariski-dense in X.

Definition 55 Let X be an (irreducible) projective variety over $\mathbf{k}$ and $f : X \dashrightarrow X$ a dominant rational map. We say that a pair (X, f) satisfies

1. *ZD-property*, if Conjecture 53(1) holds true;
2. *Strong ZD-property*, if Conjecture 53(2) holds true;
3. *AZD-property*, if Conjecture 54 holds true; and
4. *SAZD-property*, if there is a non-empty adelic open subset A of $X(\mathbf{k})$ such that for every point $x \in A \cap X(\mathbf{k})_f$, its orbit $O_f(x)$ is Zariski-dense in X.

Remark 56 Conjecture 54 implies Conjecture 53. Precisely, we have:

1. SAZD-property implies AZD-property.
2. Conjecture 54 (adelic-topology version) is stronger than the classical Conjecture 53 (Zariski-topology version). Indeed, even the hypothesis on $\mathbf{k}$ in Conjecture 54 does not cause any problem. To be precise, for every pair (X, f) over $\mathbf{k}$, there exists an algebraically closed subfield K of $\mathbf{k}$ whose transcendence degree over $\mathbb{Q}$ is finite and such that (X, f) is defined over K, i.e., there exists a pair (X_K, f_K) such that (X, f) is its base change by $\mathbf{k}$. By [72, Corollary 3.29], if (X_K, f_K) satisfies AZD-property, then (X, f) satisfies strong ZD-property.

The EMMP (cf. Theorem 25) can be applied to prove Conjecture 54 (and hence Conjecture 53) for surjective endomorphisms of (possibly singular) projective surfaces, extending the smooth case in [72].

Theorem 57 *Let $f : X \to X$ be a surjective endomorphism of a projective surface X defined over the algebraically closed field $\mathbf{k}$. Assume that $\mathbf{k}$ has finite transcendence degree over $\mathbb{Q}$ (see also Remark 56(2)). Then Conjecture 54 holds for (X, f). Precisely, either $\mathbf{k}(X)^f \neq \mathbf{k}$, or there is a non-empty adelic open subset $A \subseteq X(\mathbf{k})$ such that the forward orbit $O_f(x)$ of every point $x \in A$ is Zariski-dense in X.*

Remark 58 Even the case of normal projective surfaces of Picard number 1 is highly non-trivial, see [32, Thoerem 1.12]

We end this section by mentioning known results for Conjecture 53(1).

Remark 59

1. Conjecture 53(1) was proved by Amerik and Campana [2, Theorem 4.1] when the field $\mathbf{k}$ is uncountable.
2. Xie [72, Appendix B, joint work with Thomas Tucker] proved Conjecture 54 and hence Conjecture 53 are proved for

$$f = (f_1, \ldots, f_n) \colon (\mathbb{P}^1)^n \to (\mathbb{P}^1)^n,$$

where the f_i's are endomorphisms of $\mathbb{P}^1$; see also [5, Theorem 14.3.4.2], when f_i's are not post-critically finite, and [52, Theorem 7.16] for an endomorphism $f\colon \mathbb{A}^n \to \mathbb{A}^n$ with

$$f(x_1, \ldots, x_n) = (f_1(x_1), \ldots, f_n(x_n)),\ f_i(x_i) \in \mathbf{k}[x_i].$$

3. Xie [71, Theorem 1.1] proved Conjecture 53(1) for dominant polynomial endomorphism $f\colon \mathbb{A}^2 \to \mathbb{A}^2$.
4. Ghioca and Scanlon proved Conjecture 53(1) when X is an abelian variety and f is a dominant self-map, see [23, Theorem 1.2]. The result is further generalized to the case when X is semi-abelian by Ghioca and Satriano, see [24, Theorem 1.1]; Xie proved Conjecture 54 in this abelian variety case, see [72, Theorem 1.14].
5. When X is an algebraic surface and f is a birational self-map, Xie [72, Corollary 3.31] proved Conjecture 54 (and hence Conjecture 53); see also [4, Theorem 1.3] when X is quasi-projective over $\overline{\mathbb{Q}}$ and f is an automorphism.
6. Conjecture 54 and hence Conjecture 53 have been proved when X is a *smooth* projective surface and f is a surjective endomorphism in [72, Theorem 1.15].

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